

DIGITAL QUEUE TOKEN SYSTEM FOR LIBRARY OR CANTEEN



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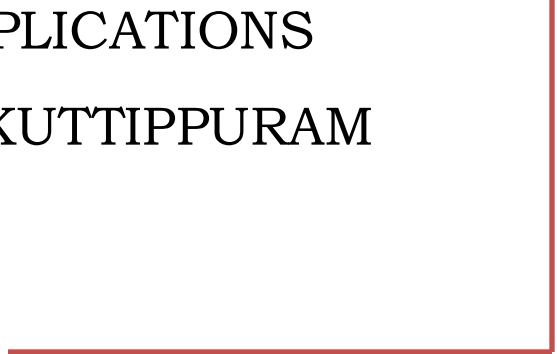


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DIGITAL QUEUE TOKEN SYSTEM FOR LIBRARY OR CANTEEN

- Digital Queue Token System helps students book tokens online for Library or Canteen, avoiding long physical queues.
- Users can see real-time queue status and get notified when their turn comes.
- Admins (staff) manage the queue by marking tokens as Completed/Skipped for smooth service

OBJECTIVES

- To reduce waiting time by providing digital tokens for Library and Canteen.
- To manage queues efficiently with real-time updates and notifications.
- To simplify admin work by allowing staff to monitor and control the queue digitally.

EXISTING SYSTEM

- Students usually stand in physical queues in library or canteen, which wastes time.
- Tokens, if used, are mostly manual paper slips without real-time updates.
- No proper system to notify students when their turn is near, leading to crowding.

PROPOSED SYSTEM

- A digital token system where students can book tokens online for library or canteen.
- Real-time queue tracking with notifications when the turn is near.
- Admin panel for staff to manage, complete, or skip tokens efficiently

MOTIVATIONS

- To save students' time by reducing long waiting in library and canteen queues.
- To avoid crowding and mismanagement by introducing a systematic digital process.
- To use technology for convenience, making queue handling smarter and more efficient

FUNCTIONALITIES

- **User Registration & Login** – Students can create accounts and log in securely.
- **Digital Token Generation** – System assigns unique token numbers for library/canteen.
- **Real-Time Queue Tracking** – Users can see their position and queue progress live.
- **Admin Queue Management** – Staff can complete, skip, or manage tokens easily.
- **Notification/Alert System** – Users get notified when their turn is near

MODULE DESCRIPTION

- **User Module** – Handles student registration, login, token booking, and queue tracking.
- **Admin Module** – Allows library/canteen staff to manage queues, mark tokens completed/skipped.
- **Queue Management Module** – Generates tokens, updates real-time positions, and maintains order.
- **Notification Module** – Sends alerts/updates to users when their turn is near.
- **History Module** – Stores previous token and queue data for records.

DEVELOPING ENVIRONMENT

- **FRONTEND** :HTML,CSS,JAVASCRIPT
- **BACKEND** :PYTHON DJANGO
- **DATABASE** :SQLITE
- **FRAMEWORK** :PYTHON DJANGO
- **IDE** :VS CODE
- **ADDITIONAL**
- **TOOL** :GIT(VERSION CONTROL)



SPRINT BACKLOG

Backlog tem	Status And Completion Date	Original Estimation in Hours	Day 1 hrs	Day 2 hrs	Day 3 hrs	Day 4 hrs	Day 5 hrs	Day 6 hrs	Day 7 hrs	Day 8 hrs	Day 9 hrs	Day 10 hrs
SPRINT 1												
User registration & Login	16/8/2025	6	1	1	0	0	0	0	0	0	0	0
Responsive Interface	22/8/2025	5	1	1	1	1	1	0	0	0	0	0
Notification system	29/8/2025	5	1	1	1	1	1	0	0	0	0	0
SPRINT 2												
Token Booking	3/9/2025	8	1	1	1	1	1	1	1	1	0	0
Real time queue display	8/9/2025	8	1	1	1	1	1	1	1	1	0	0
testing	15/9/2025	3	1	1	1	0	0	0	0	0	0	0



SPRINT BACKLOG

Backlog tem	Status And Completi on Date	Original Estimatio n in Hours	Day 1 hrs	Day 2 hrs	Day 3 hrs	Day 4 hrs	Day 5 hrs	Day 6 hrs	Day 7 hrs	Day 8 hrs	Day 9 hrs	Day 10 hrs
SPRINT 3												
Admin panel	16/9/2025	7	1	1	1	1	1	1	1	0	0	0
History and reports	20/9/2025	7	1	1	1	1	1	1	1	0	0	0
Responsive interface	26/9/2025	5	1	1	1	1	1	0	0	0	0	0
Testing	5/10/2025	1	1	0	0	0	0	0	0	0	0	0
TOTAL		55	10	9	8	7	7	4	6	4		



PRODUCT BACKLOG

Backlog ID	Feature / Requirement	Description	Estimate (Hours)
PB1	User Registration & Login	Create secure user accounts	6
PB2	Token Booking	Allow users to book digital tokens for library/canteen	High
PB3	Real-Time Queue Display	Show current queue status to users	High
PB4	Notification System	Alert users when their turn	5
PB5	Admin Panel	Manage tokens, mark complete/skip, monitor queues	7
PB6	History & Reports	Store past token and queue	6
PB7	Responsive Interface	Ensure system works on web	5



USER STORY

User Story ID	As a type of user	I want to	So that I can
US1	Student	Register and login	Access the queue system securely
US2	Student	Book a token for library/canteen	Avoid waiting in a physical queue
US3	Student	View my position in the queue	Know how many people are ahead of me
US4	Student	Receive a notification when my turn is near	Arrive on time for my turn
US5	Student	Cancel my token	Free my spot for someone else if I cannot attend
US6	Admin	View all active tokens	Manage the queue efficiently

USER STORY

User Story ID	As a type of user	I want to	So that I can
US7	Admin	Mark tokens as completed or skipped	Keep the queue updated in real-time
US8	Admin	Generate reports	Analyze usage trends for library or canteen
US9	Student	View my past bookings	Track my previous visits

PROJECT PLAN

Sprint	User Story ID(s)	Feature / Requirement	Deliverable / Task Outcome	Estimate (hrs)	Start Date	End Date
1	US1	User Registration & Login	Secure user accounts (signup, login, authentication)	6	2025-08-16	2025-08-25
1	US4	Notification System (basic)	Simple alerts/notifications for turn	5	2025-08-24	2025-09-01
1	US9 (partial)	Responsive Interface (basic)	Mobile-first UI setup	5	2025-08-29	2025-08-01
1	—	Testing & Integration	Test Sprint 1 features	1	2025-08-29	2025-09-03
Total (Sprint 1)				17		
2	US2, US5	Token Booking	Token booking + cancel option	8	2025-09-03	2025-09-12
2	US3	Real-Time Queue Display	Show live queue position/status	8	2025-09-08	2025-09-15
2	—	Testing & Refinement	Functional testing of booking & queue	3	2025-09-15	2025-09-16
Total (Sprint 2)				19		



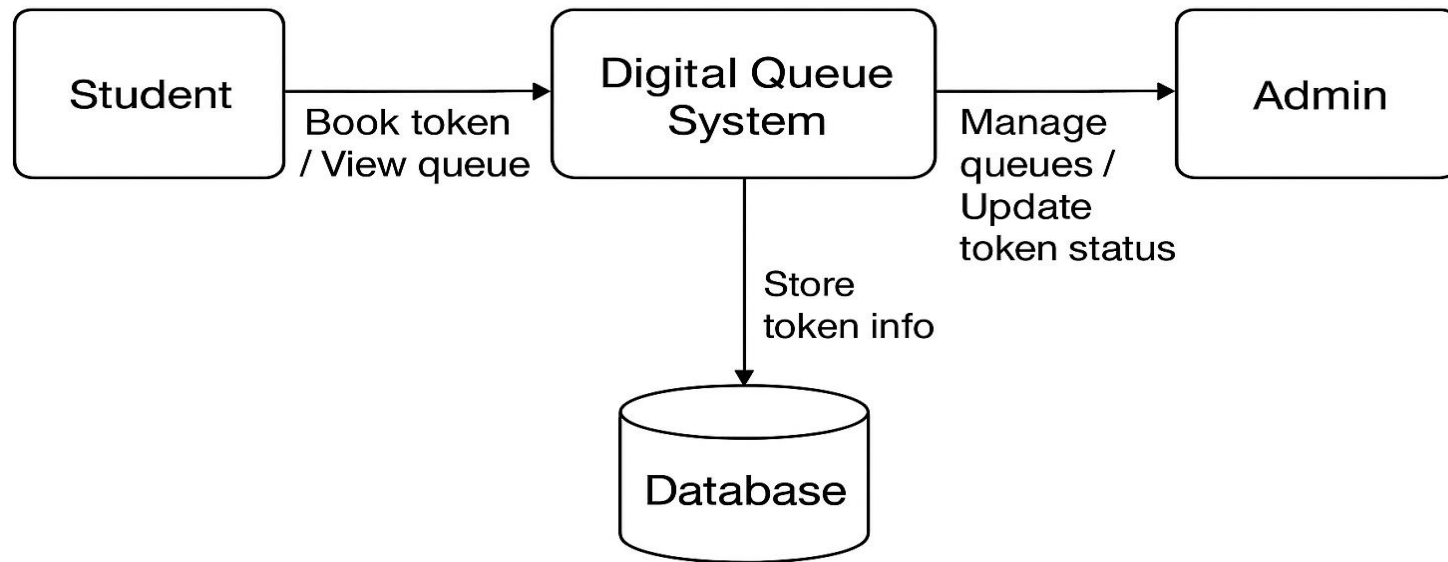
PROJECT PLAN

Sprint	User Story ID(s)	Feature / Requirement	Deliverable / Task Outcome	Estimate (hrs)	Start Date	End Date
3	US6, US7	Admin Panel	View/manage tokens, mark completed/skipped	7	2025-09-16	2025-09-25
3	US8, US9	History & Reports	Generate reports + view past bookings	6	2025-09-20	2025-09-27
3	US9 (final)	Responsive Interface (full)	Web + multi-device support	5	2025-09-26	2025-10-01
3	—	Testing & Final Integration	End-to-end testing + deployment readiness	1	2025-10-05	2025-10-07
Total (Sprint 3)				19		



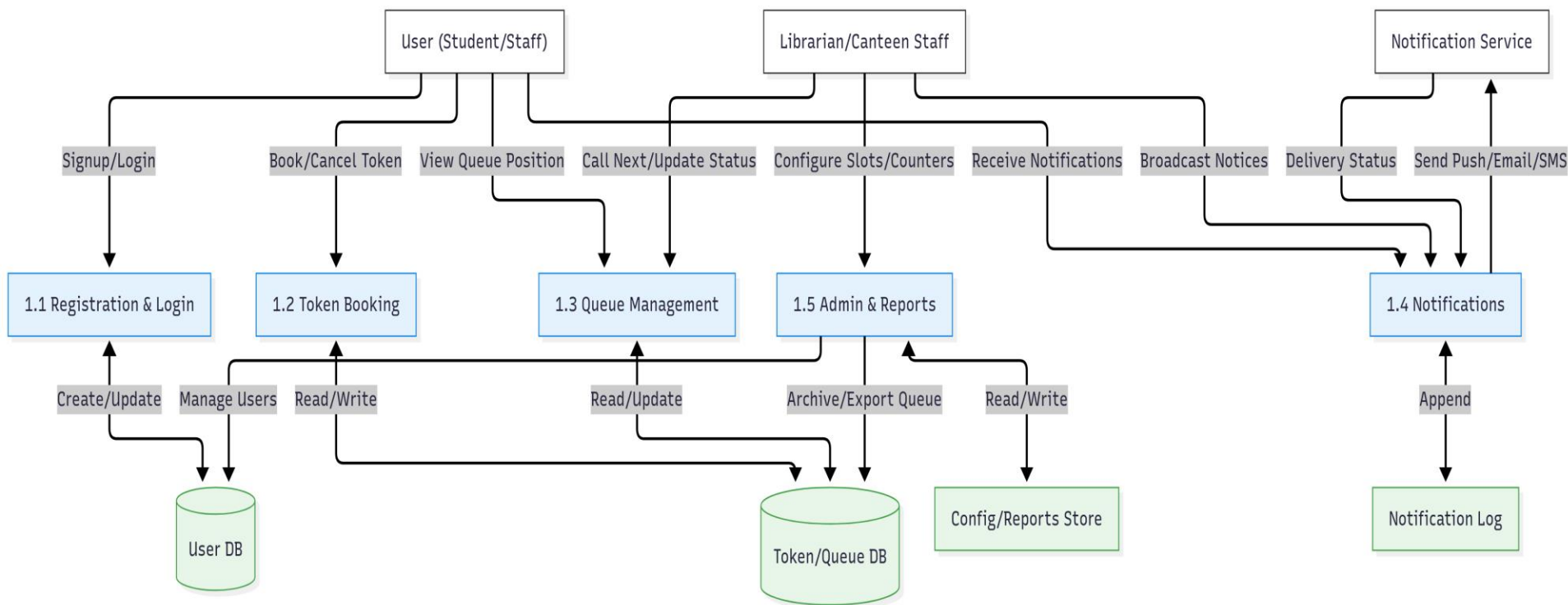
DATA FLOW DIAGRAM

- LEVEL 0



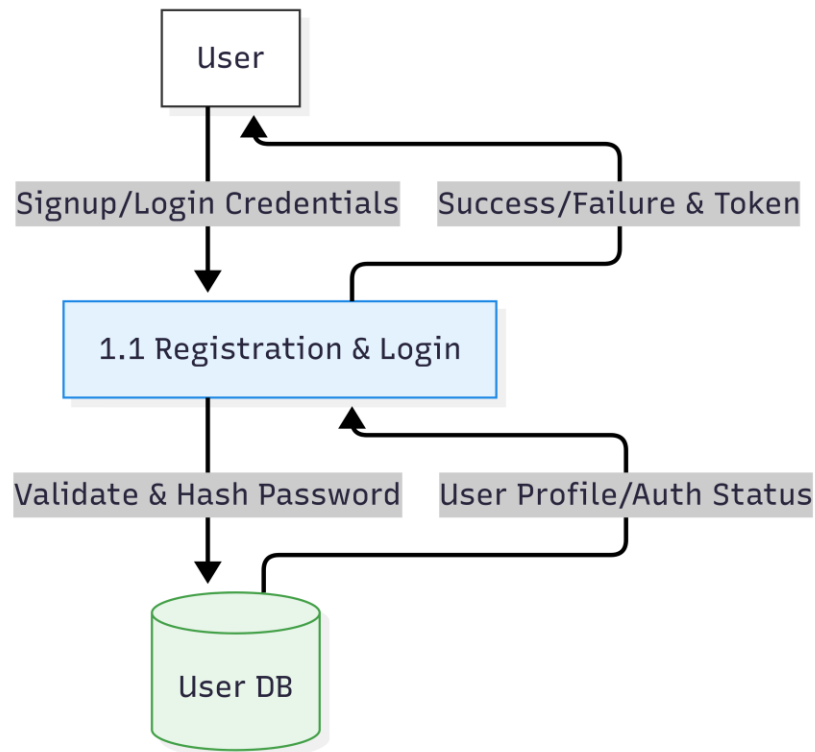
DATA FLOW DIAGRAM

- LEVEL 1



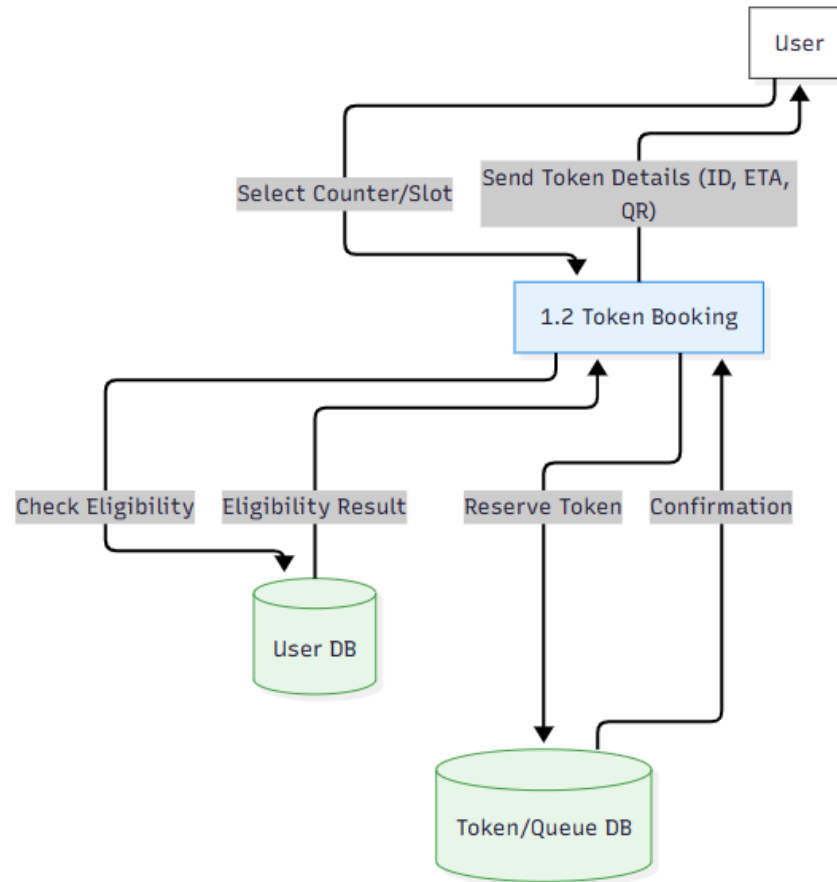
DATA FLOW DIAGRAM

- Level 1.1



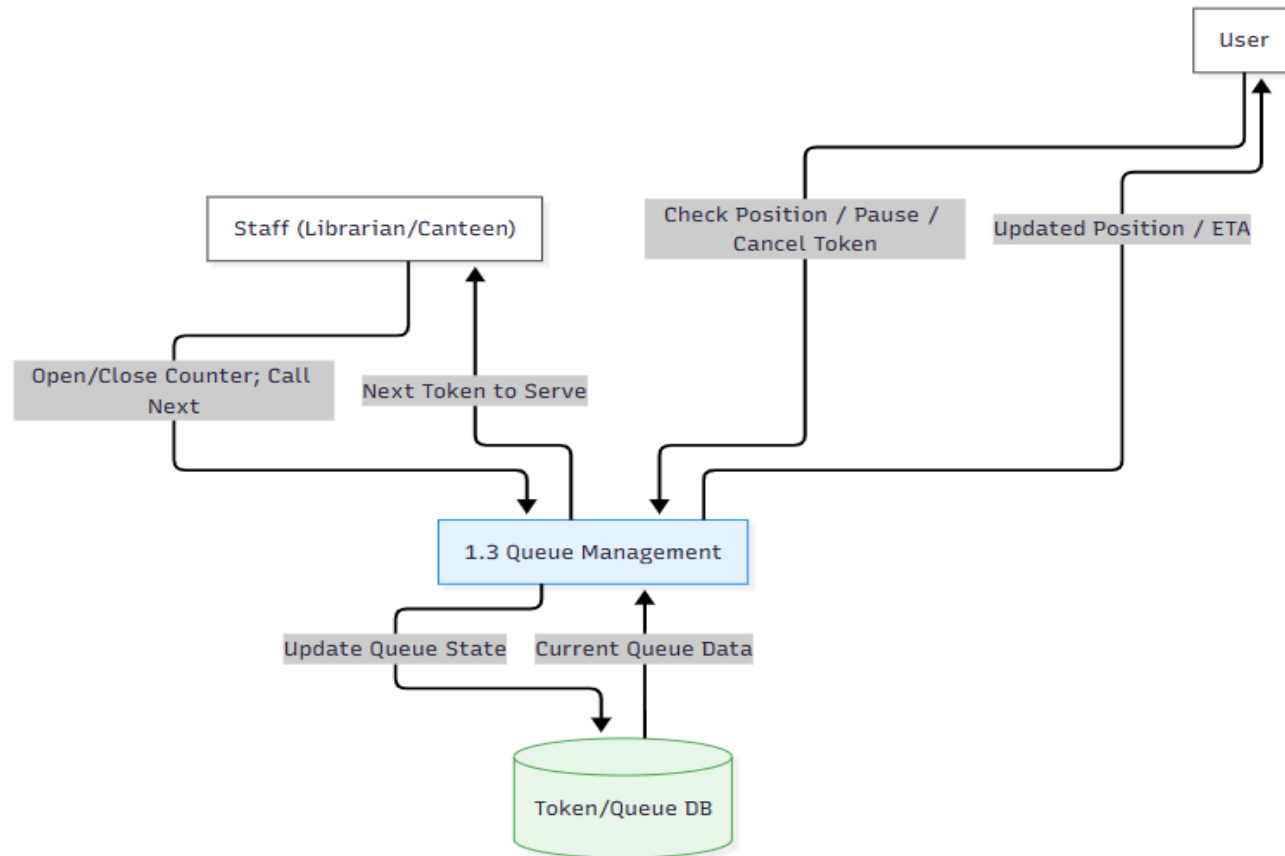
DATA FLOW DIAGRAM

- Level 1.2



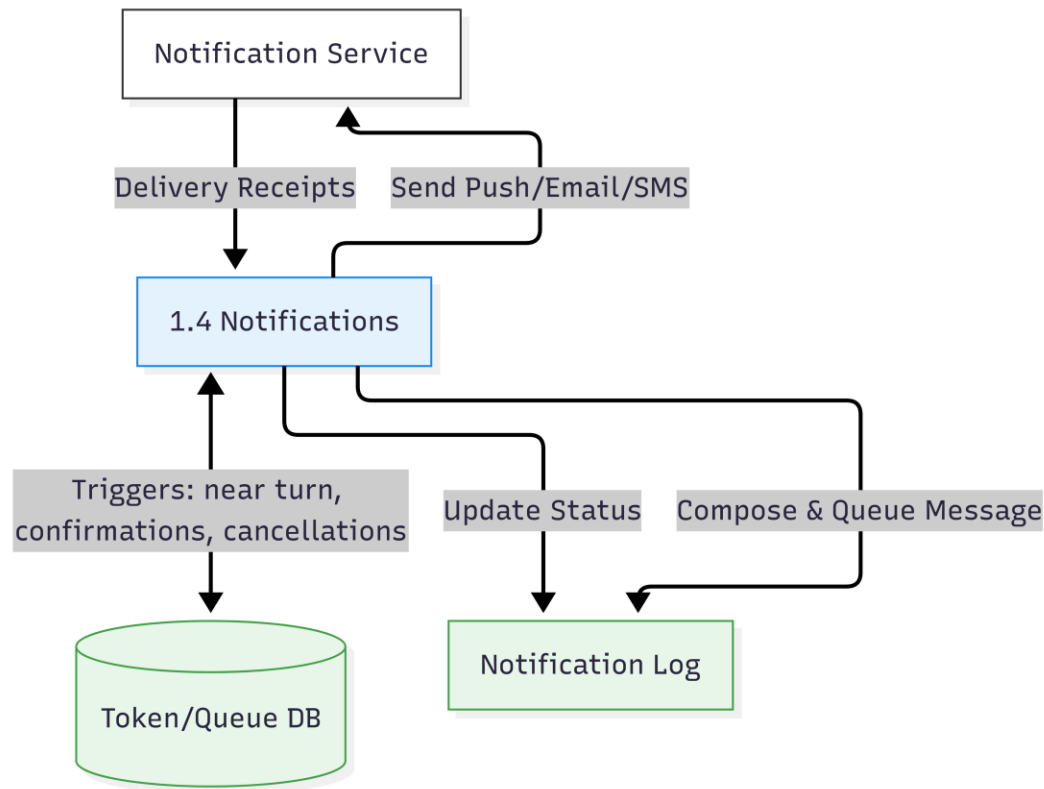
DATA FLOW DIAGRAM

- Level 1.3



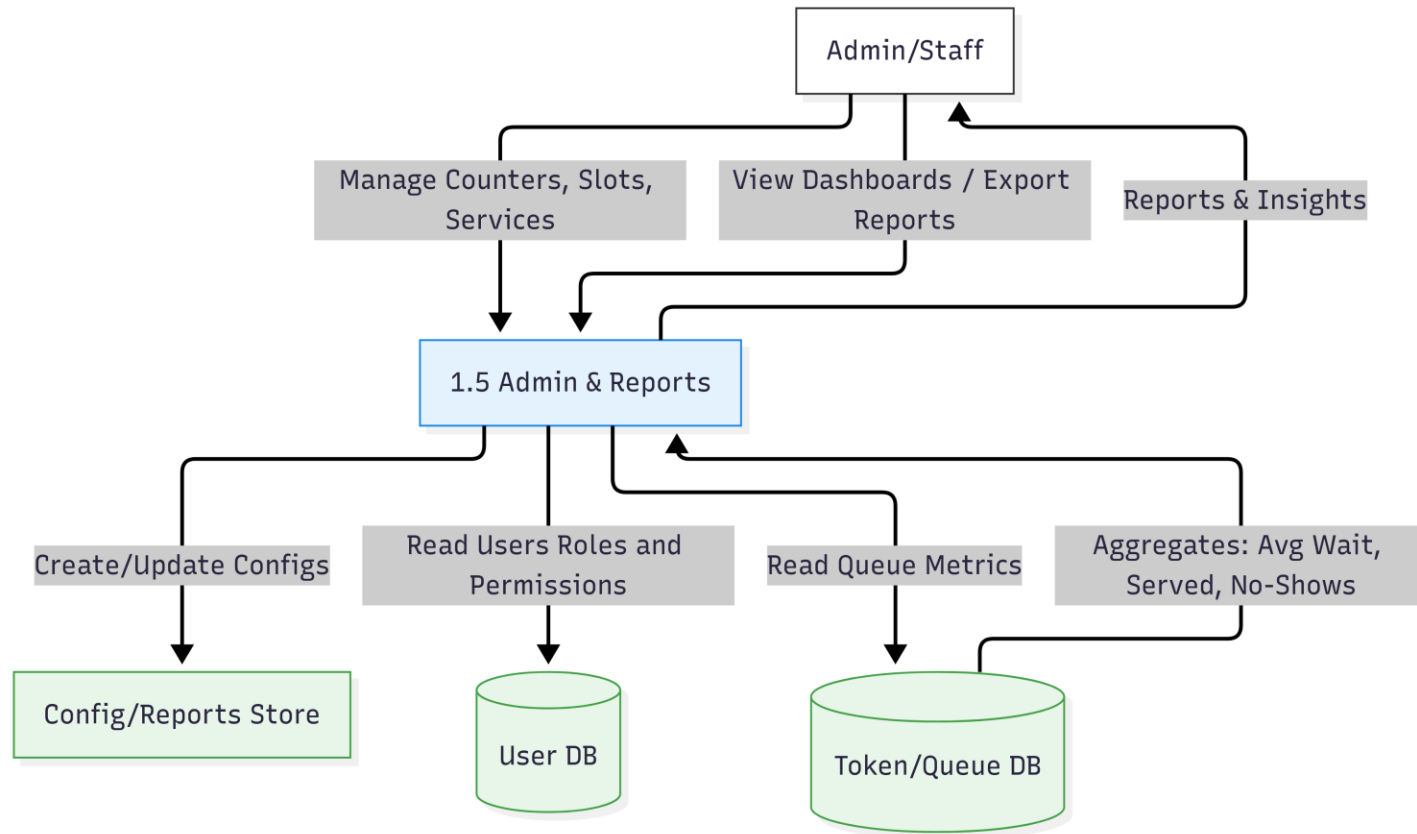
DATA FLOW DIAGRAM

- Level 1.4

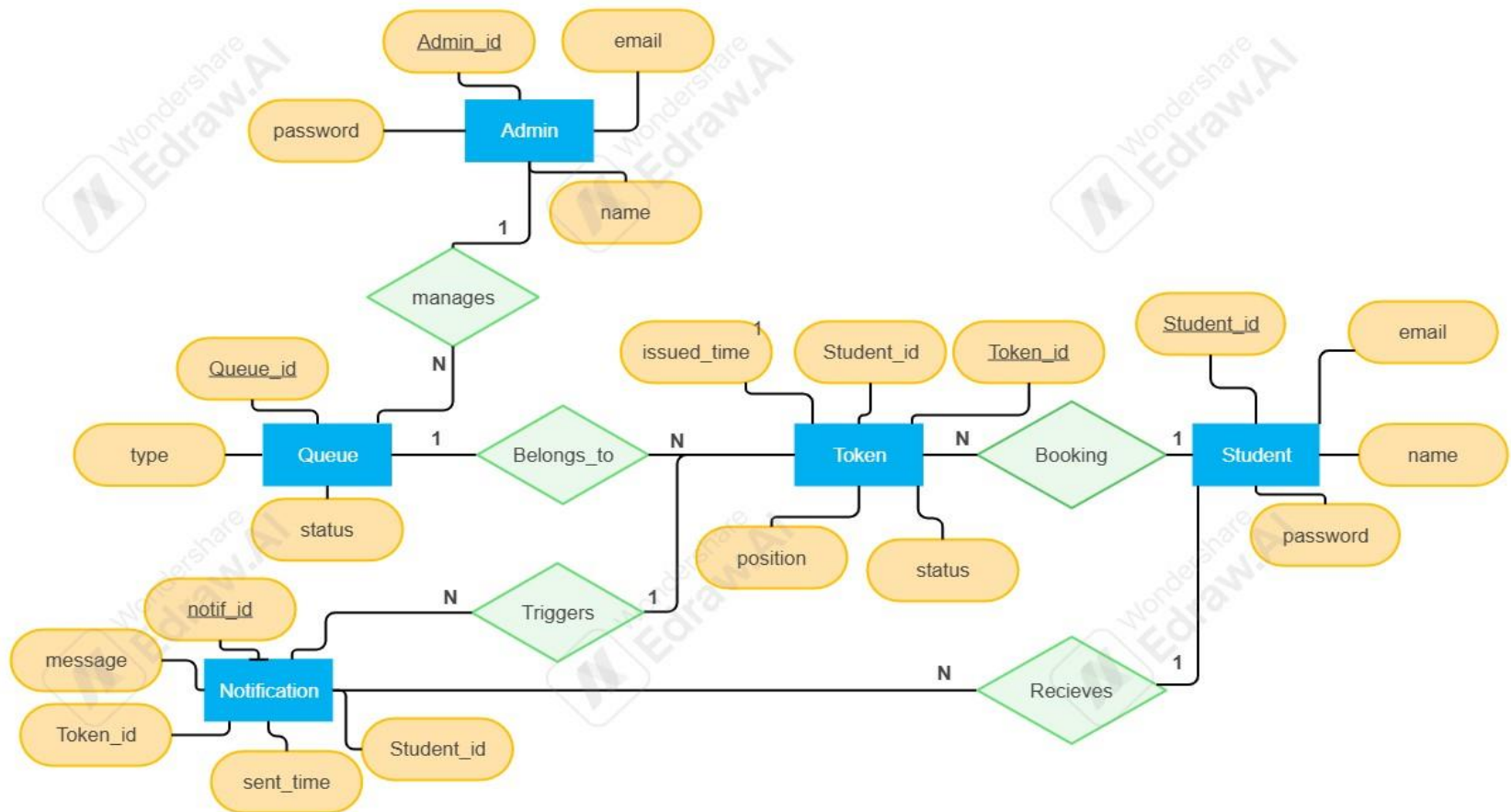


DATA FLOW DIAGRAM

- Level 1.5



ER DIAGRAM



THANK YOU

For your attention